1 Test Systems

Three power systems of different sizes and origins are described in this document: the RBTS 6-bus test system adapted from [1], the IEEE-RTS 24-bus system [2], and the synthetic Illinois 200-bus system (`ACTIVSg200`) [3]. The first is a small didactic network, used to illustrate and validate the framework in full detail; the second is a small, widely adopted reliability benchmark; and the third is a larger, realistic-scale network derived from actual grid characteristics. Together, they enable the framework to be evaluated from a small illustrative case up to a system whose size is representative of practical studies. The main characteristics of the three systems are summarized in Table 1.

1.1 RBTS 6-Bus Test System

The smallest system considered is a 6-bus test system [1]. Owing to its small size, this system is first used as an illustrative and validation case, where corrective actions and the resulting power-flow solutions can be inspected in full detail.

Because the original 6-bus test system is deterministic, stochastic reliability data were assigned in this work based on typical parameters from the IEEE-RTS 24-bus system [2]. Component failure and repair characteristics were defined according to the functional role of generators and transmission circuits within the network, with transmission elements grouped into different reliability classes reflecting their relative importance and exposure. Load uncertainty was modeled using the annual load profile from the IEEE-RTS 24-bus system, and the peak demand was increased to 120% of the original to create a more challenging operating environment with a higher frequency of corrective actions.

For this system, the GNN actor produces $6 \times 3 = 18$ candidate action components per state. After applying the action mask, voltage setpoints are adjusted at the three voltage-controlled buses (two PV buses and the slack bus); active power dispatch is adjusted at the two PV buses; and load shedding is available at the three load buses. This results in $3 + 2 + 3 = 8$ active control variables per state, while the remaining 10 components are masked out.

1.2 IEEE-RTS 24-Bus System

The IEEE-RTS, proposed in [2], is a 24-bus composite reliability test system comprising two voltage levels (138 kV and 230 kV) interconnected by transformers. It has become a standard benchmark for composite reliability assessment owing to its well-documented component reliability data.

Two scenarios are considered for this system:

- **Scenario IEEE-RTS-O (original):** the original topology, load, and generation configuration exactly as specified in the reference [2]. This scenario serves as a lightly stressed reference in which operational violations are relatively infrequent;

Table 1: Main characteristics of the test systems.

Characteristic	RBTS 6-bus	IEEE-RTS 24-bus	Illinois 200-bus
Buses	6	24	200
PQ buses	3	13	151
PV buses	2	10	48
Slack bus (SW bus)	1	1	1
Load buses	3	17	108
Circuits	11	38	245
Generating stations	3	14	49
Synchronous condenser	0	1	0

- **Scenario IEEE-RTS-R (reduced capacity):** a stressed variant in which the power flow capacities of the transmission lines and transformers are reduced to 60% of their original values, with the sole exception of the underground cable circuits, whose capacities are kept unchanged. By tightening the transmission limits, this scenario substantially increases the incidence of constraint violations and load shedding, thereby exercising the corrective capability of the GNN actor and the AC-OPF shield under more demanding conditions.

For this system, the GNN actor must produce $24 \times 3 = 72$ candidate action components per state. Applying the mask, voltage setpoints are adjusted at the 11 voltage-controlled buses (10 PV buses plus the slack bus); active-power dispatch is adjusted at the 9 generator buses that hold dispatchable active capacity (the 10 PV buses excluding the bus containing the synchronous condenser, which has no active-power generation capability); and load shedding is available at the 17 load buses. This yields $11+9+17 = 37$ active control variables per state, while the remaining 35 components are masked out.

1.3 Illinois 200-Bus System (ACTIVSg200)

The Illinois 200-bus system (`ACTIVSg200`) is a synthetic network built to reproduce the statistical and topological characteristics of a real transmission system in the region of Illinois, USA [3]. Compared with the IEEE-RTS 24-bus system, it offers a substantially larger and more realistic test case, allowing the scalability of the proposed framework to be assessed.

This system is publicly available in Matpower format [3], which provides the deterministic network data (i.e., bus, branch, and generator parameters) but does not include the stochastic component data required for composite reliability assessment. Therefore, in this work, these data were derived from the available deterministic information using typical parameter values adopted from the IEEE-RTS 24-bus system, as described below.

Transmission components were first classified by their terminal voltage levels into overhead lines and transformers, and a failure rate λ_f and a MTTR were assigned to each. For overhead transmission lines, the failure rate was modeled as length-dependent,

Table 2: Stochastic data - transmission components - Illinois 200-bus system.

Component	**Quantity**	λ_f **(f/year)**	**MTTR (h)**
Overhead transmission line, 230 kV	21	$0.0034\,\ell + 0.29$	11
Overhead transmission line, 115 kV	158	$0.0052\,\ell + 0.22$	10
Transformer, 13.8/115 kV	34	0.02	768
Transformer, 13.8/230 kV	15	0.02	768
Transformer, 115/230 kV	17	0.02	768

$\lambda_f = a\,\ell + b$ (in failures per year), where the per-mile coefficient a and the base term b differ by voltage level. Since the physical line lengths are not provided in the original data, the length ℓ (in miles) of each circuit was inferred from its per-unit series reactance x as $\ell = x/x_{\text{pu/mile}}$, using a typical reactance-per-mile constant $x_{\text{pu/mile}}$ for each voltage class: 0.00144 pu/mile at 230 kV and 0.00384 pu/mile at 115 kV. Transformers, identified as branches connecting two distinct voltage levels, were assigned a fixed failure rate of $\lambda_f = 0.02$ f/year and a long repair time of MTTR = 768 h, reflecting their low failure frequency but time-consuming repair. The resulting line and transformer reliability parameters are summarized in Table 2.

A caveat is considered: circuits that radially connect terminal load and generation buses are excluded from the contingency set (23 circuits in this system). The outage of such a circuit inevitably disconnects its associated load bus, resulting in load curtailment that is entirely determined by the network topology. Since this interruption is independent of the operating state, it can be predicted a priori without requiring power-flow or optimization analyses. Including these contingencies would introduce deterministic, topology-driven contributions to the reliability indices that provide no information about the adequacy of the meshed portion of the network, which is the focus of the present composite reliability assessment. Moreover, such contingencies correspond to corrective states for which only one feasible response exists: complete load shedding at the disconnected bus, which neither the AC-OPF nor the proposed GNN-based corrective policy can avoid. For these reasons, the reliability indices reported for this system exclude load-shedding events associated with outages of radial terminal circuits. The resulting indices should therefore be interpreted as measures of the reliability of the composite meshed network rather than of the entire distribution topology.

Generating units were grouped into technology classes according to their primary-mover type, and each class was assigned a representative λ_f and MTTR. Four classes are present in the system: steam turbine (coal), wind, gas turbine, and nuclear. Their reliability parameters and the number of units in each class are listed in Table 3.

To simulate an operating condition representative of a planning context, such as a transmission expansion study in which the system is deliberately driven toward its operational limits, a single stressed scenario is considered. In this scenario, the active and reactive loads at all buses available in [3] are treated as peak demand and increased to 150% of their original values. The uncertainty of the load is then modeled using the yearly load curve in [2]. In addition, the power flow capacities of the transmission lines

Table 3: Stochastic data - generating units - Illinois 200-bus system.

Generation class	**Quantity of Units**	λ_f **(f/year)**	**MTTR (h)**
Steam turbine (coal)	25	4.47	40
Wind	6	5.00	30
Gas turbine	17	19.47	50
Nuclear	1	7.96	150

and transformers are reduced to 90% of their original capacities. Furthermore, all capacitor banks providing reactive compensation are deactivated, as fixed compensations can introduce severe voltage regulation difficulties under fluctuating load conditions (if these assets were instead treated as controllable, their corresponding buses would be modeled as PV buses, allowing voltage magnitudes to be adjusted through the actions of the RL agent). The simultaneous increase in demand, reduction in transmission capacity, and absence of fixed reactive support emulate a future, more heavily loaded operating condition. This combination results in a higher frequency of operational violations, thereby providing a stringent test of the framework on a large-scale network.

In this case, the GNN actor produces $200 \times 3 = 600$ candidate action components per state. Applying the mask, voltage setpoints are controlled at the 49 voltage-controlled buses (48 PV buses plus the slack bus); active-power dispatch is controlled at the 48 PV buses, all of which hold dispatchable active capacity; and load shedding is available at the 108 load buses. This yields $49+48+108 = 205$ active control variables per state, with the remaining 395 components masked out.

References

[1] A. J. Wood and B. F. Wollenberg, *Power Generation, Operation, and Control.* New York, NY, USA: John Wiley & Sons, 2 ed., Jan. 1996.

[2] PMS, "IEEE Reliability Test System," *IEEE Trans. Power Appar. Syst.*, vol. PAS-98, no. 6, pp. 2047–2054, 1979.

[3] A. B. Birchfield, T. Xu, K. S. Shetye, and T. J. Overbye, "Activsg200: Synthetic 200-bus power system test case." `https://electricgrids.engr.tamu.edu/electric-grid-test-cases/activsg200/`, 2018. Accessed: 2026-06-03.